

F O R T I M A N A G E R N S E 6

Lab 03 — Concurrent Administration

01. Session Monitoring

FortiManager is a shared system by design. Several administrators work on the same appliance, often inside the same ADOM, and the first question before changing anything is who else is connected.

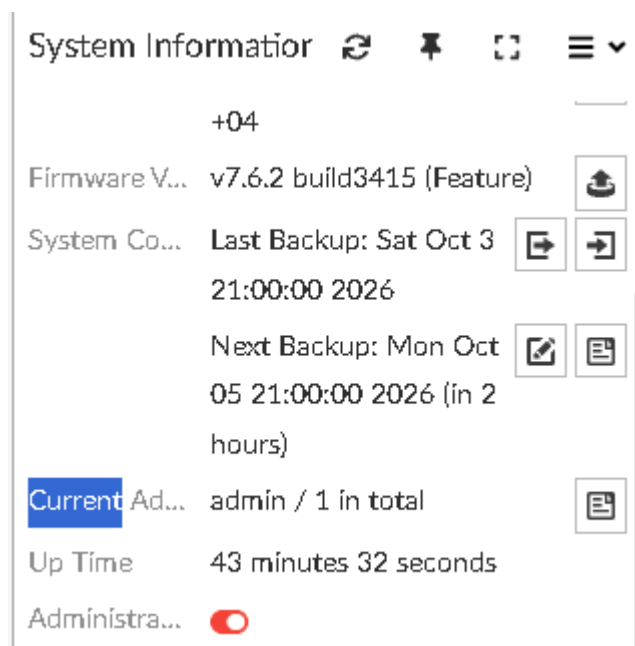


Figure 1 — The System Information widget reporting the current administrator count.

The count on the dashboard is a summary. The session list behind it is the detail: account, profile, source address, which ADOM the session is working in, when it started, and how long it has been idle.

	User Name	Profile	IP Address	Current ADOM	Start Time	Idle
☐	A: admin (Current)	Super_User	GUI/172.16.28.147	root	Mon, Oct 5, 2026 06:24:48 PM	16s

Figure 2 — The Admin Session List — account, profile, source, current ADOM, start time and idle time.

```
diagnose system admin-session list
```

```

FortiManager-Laps # diagnose system admin-session list
*** entry 0 ***
session_id: 57010 (seq: 0)
username: admin
admin template: admin
from: GUI(172.16.28.147) (type 1)
profile: Super_User
adom: root
session length: 135 (seconds)
idle: 22 (seconds)

*** entry 1 ***
session_id: 39053 (seq: 1)
username: admin
admin template: admin
from: console (type 0)
profile: Super_User
adom: root
session length: 6 (seconds)

```

Figure 3 — The same information from the CLI, including the session ID.

The CLI output distinguishes connection types that the GUI shows less plainly. (type 1) is a GUI session with its source address; (type 0) is the console. Both appear under the same account here, which is worth noting: **one administrator can hold several concurrent sessions, and each holds locks independently of the others.**

Sessions can be deleted from the list. On an appliance using workspace locking this is the practical recovery path when a lock is held by an abandoned session — covered in section 3.4.

02. Workspace Modes

By default FortiManager applies no locking. Two administrators can open the same object, edit it, and save — and the second save overwrites the first with no warning to either of them. **Nothing in the interface indicates that a change was lost.** Workspace mode is the mechanism that prevents this.



Figure 4 — System Settings → Advanced → Workspace, showing the four available modes.

Mode	Behaviour
Disable	No locking. The default, and the behaviour described above.
Workspace (ALL ADOMs)	Locking enforced across every ADOM on the appliance. An administrator must lock before editing, then commit or discard.
Workspace (Per-ADOM)	Locking enforced only in ADOMs where it is enabled. Allows a staged rollout, or a mixed estate where one tenant requires change control and another does not.
Workflow (ALL ADOMs)	Locking plus an approval cycle. Changes are submitted rather than applied, and take effect only once an approver accepts them.

► 2.1 Per-Policy Lock



Figure 5 — Workspace enabled across all ADOMs, with Per-Policy Lock available.

Without it, locking an ADOM locks everything inside it — one administrator editing one policy package blocks every other administrator from touching anything in that ADOM. Per-Policy Lock narrows the granularity so that individual policy packages are locked instead.

On a large ADOM with several teams working in parallel this is the difference between a workable system and one where administrators queue. The trade-off is that object-level changes still require the wider lock, since objects are shared across packages.

► 2.2 Changing the mode disrupts active sessions

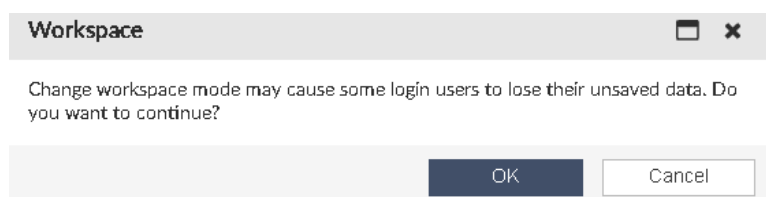


Figure 6 — The warning shown when the workspace mode is changed.

Administrators with unsaved work lose it when the mode changes. FortiManager rebuilds its session state around the new model, and anything held in a session that has not been committed does not survive. On a production appliance this is a scheduled change, not an ad-hoc one.

03. Locking

Once workspace mode is enabled, everything is read-only until an administrator takes a lock. The ADOM list gains a lock indicator per ADOM, and the banner shows the state of whichever ADOM the session is working in.



Figure 7 — The ADOM list with workspace enabled. Each ADOM carries its own lock state.

► 3.1 Reading the lock indicator

Indicator	Meaning
Grey, open padlock	Unlocked. The ADOM is readable by anyone and editable by nobody.
Green, closed padlock	Locked by this session. Editing is permitted, and changes are staged until committed.
Red, closed padlock	Locked by a different administrator. Read-only for this session.



Figure 8 — Unlocked — the ADOM is read-only for everyone.



Figure 9 — Locked by this session — editing permitted.



Figure 10 — Locked by another administrator — read-only for this session.

The red indicator is the single most useful diagnostic in FortiManager. An administrator reporting that “everything is read-only” is almost always looking at it without recognising what it means.

► 3.2 Attempting to lock a held ADOM

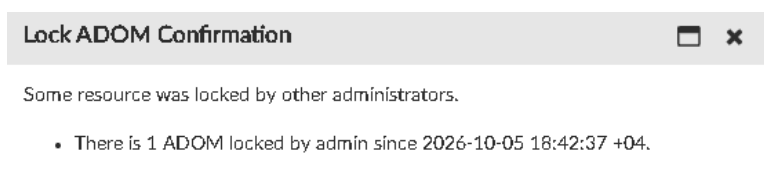


Figure 11 — The lock attempt refused, naming the holder and the time the lock was taken.

The refusal is specific: which resource is held, by whom, and since when. That is enough to resolve the situation without guesswork — contact the holder, or, if the session is abandoned, delete it from the session list in section 1.

► 3.3 Editing under a lock



Figure 12 — A configuration change made while the lock is held.

A change made under a lock is staged rather than applied. It exists in the administrator's workspace and is invisible to every other session until it is committed. Three outcomes are possible:

- **Save** stores the change in the workspace. It is not yet live and the lock is retained.
- **Commit** applies the staged changes and releases the lock.
- **Discard** throws the staged changes away and releases the lock. The object returns to its pre-lock state.

This is the real purpose of workspace mode, and it is not only about collision avoidance. An administrator can make a set of related changes, review them as a whole, and apply or abandon them atomically — rather than each edit taking effect the moment it is saved.

► 3.4 Lock preemption

```
Fortimanager-Lapss # config system global
(global)# set lock-preempt enable
(global)# end
```

Figure 13 — Enabling lock preemption.

```
config system global
    set lock-preempt enable
end
```

Preemption allows an administrator to take a lock held by another. The case for it is practical: a lock taken at the end of a working day and not released blocks the ADOM until the holder returns, and on a 24-hour operation that is not acceptable.

Preemption discards the other administrator's staged work

Taking a held lock does not merge the two sessions. The preempted administrator's uncommitted changes are lost, exactly as in the warning shown when the workspace mode itself is changed. Enabling preemption trades a reliability problem for a data-loss one, and the choice depends on which is worse in the environment.

The alternative — deleting the stale session from the Admin Session List — has the same effect on the staged work but leaves the preemption setting off, so it cannot happen by accident during normal operation.

04. Workflow Mode and Change Acceptance

Workflow mode extends locking with an approval cycle. An administrator locks, edits, and *submits* rather than commits. The change sits pending until an approver accepts or rejects it, and only on acceptance does it take effect.

Stage	What happens
Lock	As in workspace mode — the administrator takes the ADOM or policy package.
Edit and save	Changes are staged in the session, invisible to others.
Submit	The session is sent for approval with a description of the change. The administrator cannot apply it themselves.
Approve or reject	An approver reviews the session. Approval applies the changes; rejection returns them to the submitter.

The separation is the point: the person who makes a change is not the person who authorises it. In a regulated environment, or anywhere a firewall change requires a change-control record, that separation is the requirement rather than a convenience.

► 4.1 Changes received from the FortiGate

Configuration Changes Received from FortiGate

Automatically accept Prompt Administrator to accept

Figure 14 — The handling of configuration changes originating on a managed FortiGate.

Setting	Behaviour
Automatically accept	A change made directly on the FortiGate is accepted into FortiManager's device database and recorded as a new revision without administrator involvement.
Prompt Administrator to accept	The change is held and an administrator must decide whether to take it. FortiManager and the device remain out of sync until that decision is made.

This setting is the inbound counterpart to workflow approval. Workflow governs changes flowing from FortiManager to devices; this governs changes flowing the other way. An organisation that has gone to the trouble of requiring approval for outbound changes, while silently accepting every change made at the device CLI, has only closed half the loop.

The trade-off is familiar. Automatic acceptance keeps the device database current with no effort and removes the record of who changed what outside the management system. Prompting preserves that record at the cost of a queue somebody has to work through.

05. Trusted Hosts

Trusted hosts restrict where an administrator account may connect from. The account authenticates normally, but a login attempt from outside the permitted ranges is refused regardless of the password.

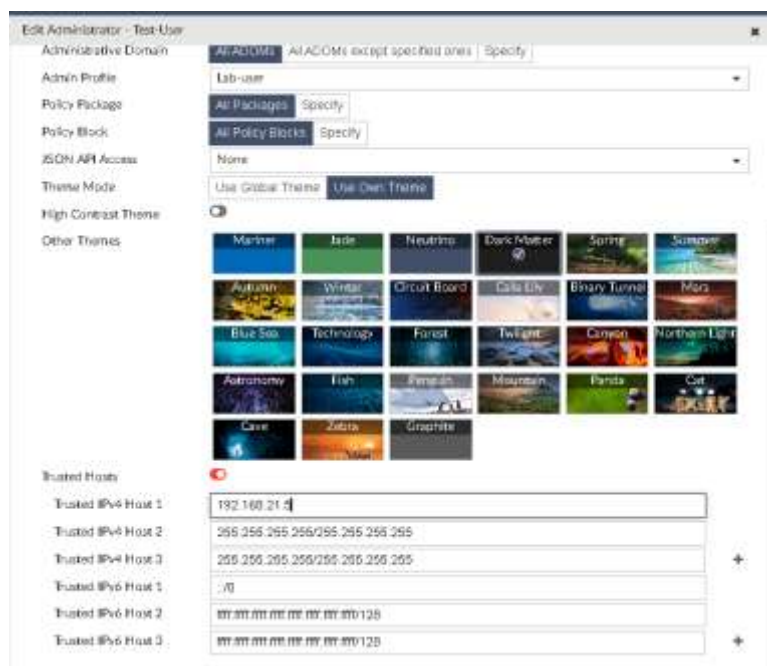


Figure 15 — Trusted hosts enabled on the Test-User account, restricted to a subnet that excludes the test workstation.

FortiManager-VM64

Authentication failure. Please try again...

Test-User

...

Login

Figure 16 — The login refused, as expected.

Observation	The refusal is reported as Authentication failure. Please try again — the same message an incorrect password produces.
Why it matters	The interface does not distinguish a wrong password from a disallowed source address. An administrator locked out by a trusted-host entry will report the problem as a password issue, and will keep reporting it until someone checks the account configuration.
Diagnosis	The Event Log records the source address of the refused attempt, which is what identifies the real cause.

Caution

The default entry is `0.0.0.0/0` — all sources. Replacing it with a specific host while connected from a different address locks the account out immediately, including the account making the change if it is applied to the wrong user.

On a FortiManager the control is worth more than it would be elsewhere. An administrator account here holds authority over every managed firewall, so limiting the account to the management subnet reduces what a stolen credential can reach.